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Research paper



On characterization of a generic lithography machine in a multi-directional space

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ABSTRACT

In this paper, we present a dynamics modeling approach for lithographic machines. These machines consist of N-open chains with multiple stages in each chain. A stage can suppress or exert motion in specific directions. The stages affect each other through the structural connections available within a chain or various chains. We want to capture the energy transfer in the machine to better realize the design requirements. A specific application will be the optical lithography wafer-scanner machine. In such machines, step-and-scan motion profiles demand synchronization between the chains end-effectors. The rapid production rates requirements are projected onto the design space as high-acceleration profiles that will cause vibration based on the energy paths available. Various positioning errors will take place during the production of one complete substrate. Characterizing the machine dynamics due to jointly motion and vibration helps in providing better ways to synthesis the controllers for each stage. To minimize the positioning errors, we also propose replacements of specific stages where actively controlled fine positioning stages are used. The presented machine characterization provides a means to design and optimize such stages as will be illustrated through simulation examples.

1. Introduction

In semiconductor manufacturing, optical lithography machines are used to produce nano-metric integrated circuits (ICs). As depicted in Fig. 1, these machines use an optical system to print integrated circuits images onto a silicon disk, known as *wafer*, which is coated with a light sensitive material [1]. The integrated circuit pattern (the mask) exists of a quartz plate, known as *reticle* [2], which is a part of the reticle stage motion control system. In these machines, step-and-scan techniques are adopted, where a small rectangular area of the wafer, known as the exposure field or *die*, is exposed (scanning process) during which the wafer is moving with constant velocity – by the wafer stage – in a direction opposite to the reticle's motion direction. Once the projection of a single die is done, the wafer stage moves the wafer to the next die location (stepping process) [3].

Driven by the increased demand of higher throughput and yield, the lithography machine is required to withstand extreme inertial forces caused by the high acceleration values, which may exceed 120 m/s^2 for the reticle stage, and 45 m/s^2 for the wafer stage during stepping motion [2,4]. The resulting inertial forces induce undesired vibrations in the lithography machine [4], and may cause reticle-slip phenomena [5]. The induced vibrations in lithography machines can deteriorate the required nanometric positioning accuracy, and significantly reduce the machine yield [2]. Consequently, these vibrations prevent the lithography machine

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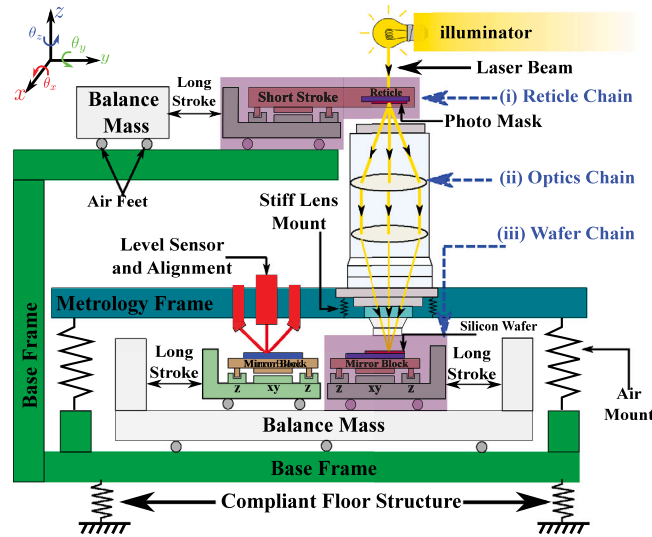


Fig. 1. A schematic representation of the lithography machine architecture, which consists of three main parts: (i) the reticle chain, (ii) the projection lenses, and (iii) the wafer chain. The x and y -directions represent the stepping and scanning directions, respectively.

from reaching the desired critical dimension (CD) and the specification on overlay. Moreover, the induced vibrations can affect the optics system and cause leveling errors which lead the silicon wafer to be out of the lens focal plane [2].

One way to minimize the static leveling errors is by taking offline maps of the wafer surface topology before the exposure takes place, as can be found in TWINSCANTM machines, [2]. Based on the obtained wafer surface topology, leveling of the wafer stage can be done by trading the wafer set-point via feedforward while jointly suppressing the vibrations experienced by the optics system. In [6], the small magnitude and long duration vibration is distinguished from the large magnitude and relatively short duration shock excitation. Indeed, both types of vibration must be considered to have a robust opto-mechanical system.

1.1. Existing literature

Since the positioning accuracy of the reticle and the wafer is crucial to produce efficient ICs [7], minimizing the micro-positioning errors as well as the point-of-interest, as at the point-of-control, is considered as the main performance requirement in lithography process [8]. Several studies contribute to enhance the lithography process. Some of these contributions are focused on the control aspect of the machine, as in [1,9–13], and others are focused on introducing novel devices (or mechanisms) to improve the performance of this machine, as in [14–16].

The error modeling is the most important and critical step in error analysis in precision motion. Kinematic chain and homogeneous coordinate array methods are the most used approaches to model the dynamical errors in these machines.

The kinematic chain and homogeneous coordinate array approaches have been utilized in modeling the errors in lithography machines. For example, the lithography machine has been designed and optimized in [17] through an error analysis model for the 6-DOF of the reticle stage.

In addition, in [18], a detailed model of a wafer stage position-dependent behavior is obtained using modal analysis, where the applicability of the proposed method to enhance the control of high-precision motion systems is demonstrated through simulation and experiment. Furthermore, in order to obtain an accurate reference position of the reticle stage, a 6-DOF forward and inverse kinematic analyses have been derived in [19], where the overall error of the reticle stage has been reduced, and the accuracy of the reticle stage has been improved.

Based on the previous discussion, it is obvious that the positioning accuracy of the reticle and the wafer stages is very critical to produce efficient ICs [7]. Also, the synchronized motion between the reticle and the wafer stages is essential to avoid the focus and overlay errors. Hence, the main performance requirement in lithography process is to minimize the relative positioning error between the two stages in all 6-DoF for each of the individual stages [8].

In [14], a reticle curvature manipulator is proposed to reduce the focus error due to lens heating and wafer unflatness. The proposed manipulator implements *fourteen* piezoelectric actuators through two arrays of actuation mechanisms located at the edges of the reticle. These actuators provide the reticle with a uniform time varying bending moment, as curvature correction technique, resulting in an estimated focus error reduction of approximately 10 nm.

A novel reticle clamping concept, that allows enhanced scanning accelerations, is investigated in [15]. The proposed clamping system eliminates the reticle slip by avoiding acceleration force transfer by means of friction, and reduces the non-correctable reticle deformations by employing struts to the reticle to constrain it. In [16], reticle slip is eliminated by proposing a reticle assist device which consists of a coarse stage, for accommodating reticle load/unload, and a fine stage which utilizes piezoelectric stack capable to exert a controlled pushing force on the reticle and cancel the inertial loads.

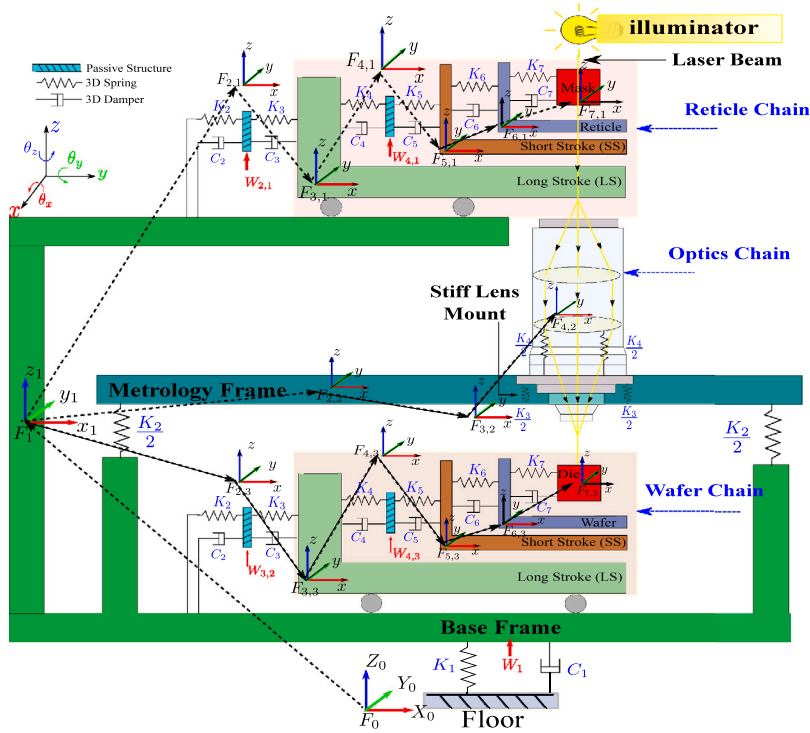


Fig. 2. A lumped mass-spring-damper system representing the open-loop chains of the lithography machine. The externally applied i th wrench in the j th chain is denoted by $W_{i,j}$.

1.2. Study objectives

This paper is divided into two parts, the first one mainly proposes a comprehensive *low-fidelity model* that is suitable to study motion-related dynamics of the lithography machine considering the 6-DOF of the positioning stages *based on Newton's and Euler's equations*. The model presented herein can be used for performance analysis and control design of specific stages and/or the whole machine. Also, it helps in capturing the errors due to floor vibrations, projection optics box vibrations, and the coupling between different stages within the machine and by extension its neighboring machines all in 6-DOF. In this study, we focus mainly on positioning-related errors among which chains synchronization error is of primary interest. The second part proposes a generalized lumped mass-spring-damper model of fine positioning stages comprising a multitude of actuators. This novel *idea* of introducing one or more fine positioning stages with the needed degrees of freedom allows compensating positioning errors, and hopefully to enhance the machine throughput. These additional stages will make the positioning in the machine multi-stage dependent instead of being mainly dual-stage dependent. The effect of introducing the fine stages is then studied with the aid of the developed machine model. During the step-and-scan motion, a performance comparison between a machine equipped with a fine positioning stage and a machine equipped with a conventional reticle stage design is studied through simulation. Also, the case of having two fine stages at different locations to regulate positioning errors in 6-DOF is investigated. The mathematical formulation of the lithography machine dynamics model is given in Section 2, modeling the fine stage is presented in Section 3, open-loop chains interactions is covered in Section 4, positioning-related error is covered in Section 5, and the error correction is assessed in Section 6. Final remarks are provided in Section 7.

2. Dynamics modeling of the lithography machine

A rigid multi-body system that consists of discrete rigid bodies (or stages) having relative motions with respect to each other can be approximated as a chain [20]. By definition, when a chain has one base-body and one end-effector, and when each stage has one parent and one child, then its topology is serial [20]. Since a serial-chain has one end-effector, it forms an open-loop chain. Fig. 4 shows a typical actuated flexible serial-chain. Despite being simple, the concepts related to serial-chains can be extended to more complex and closed-loop multi-body dynamical systems [20]. Introducing the concept of chains helps in characterizing any machine, i.e. the lithography machine, as it can be treated as one multi-body system [21].

With reference to [24], mass-spring-damper systems can be used to model the mechanics in precise positioning systems. Hence, the lithography machine can be simplified and represented as shown in Fig. 2. Focusing on the positioning stages and excluding mainly the handlers, illuminator, immersion (or cooling) hood, it is possible to deal with a typical lithography machine as a machine

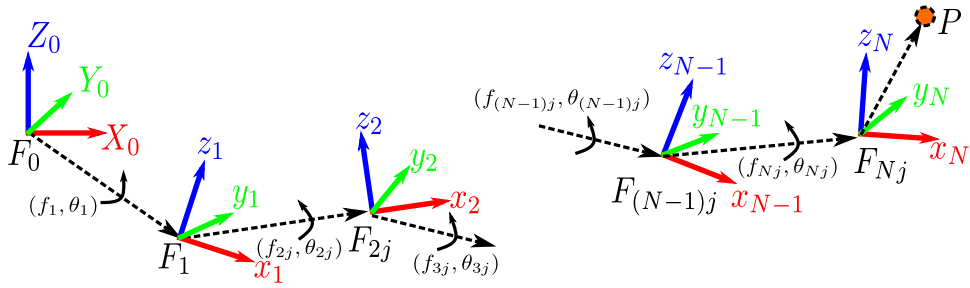


Fig. 3. The body-fixed frames attached to the CoG of their corresponding stages in a typical j th chain, where the kinematics of the point P – representing the chain end-effector – are used to assess the performance of the lithography machine.

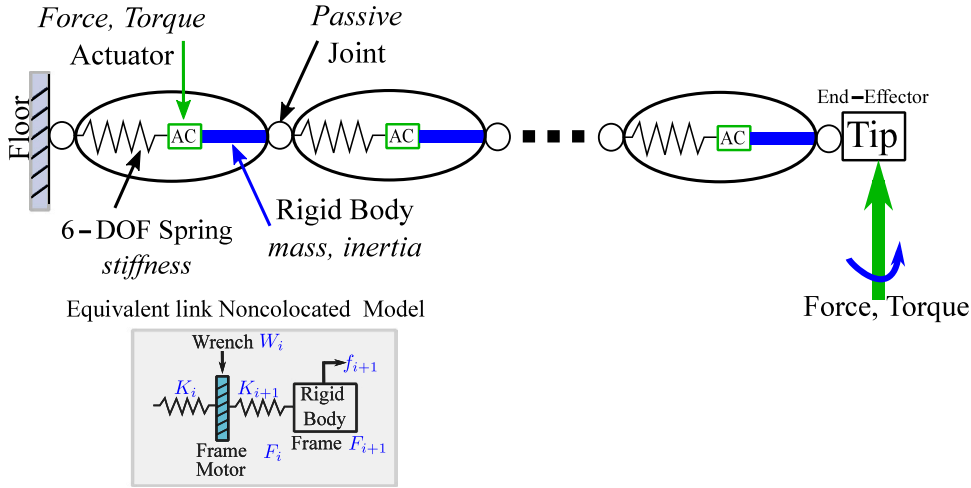


Fig. 4. An actuated flexible serial-chain with equivalent link noncolocated model where a compliant connection connects the motor and the rigid body, i.e., the stage, in all degrees of freedom [22,23].

that consists of three open-loop chains: the reticle chain depicted in Fig. 5, the wafer chain depicted in Fig. 6, and the optics chain depicted in Fig. 7. The motion of the three end-effectors, i.e. the mask, lens element – specifically the last lens element in Deep Ultraviolet machines that is called Wet Exchangeable Last Lens Element (WELLE) –, and die, must be synchronized during the scanning process. In this study, both the synchronization between the mask and the die, see Figs. 5 and 6, and also among the three end-effectors, see Fig. 5 through Fig. 7, will be investigated.

With reference to Fig. 3, it has been assumed that each chain in the machine consists of N_j stages, and therefore the body-fixed frames associated with each stage in a typical j th chain can be illustrated as shown in Fig. 3, where each frame is attached to the CoG of its corresponding stage. The externally applied i th wrench in the j th chain is denoted by $W_{i,j}$ with $i \geq 2$ and $j = 1, 2, 3$. Note that the wrench W_1 is used mainly to suppress the vibration in the whole machine, while the other wrenches are mainly used for positioning within a single chain.

As depicted in Fig. 5, a noncolocated – see Fig. 4 – dual-stage structure is adopted in the reticle chain where large range micron-scale movements are accomplished by the long-stroke stage using commutating actuators capable of strokes up to 1 meter. While the small range nano-scale dynamic tracking and positioning are conducted by the short-stroke stage using Lorentz (nowadays variable reluctance) actuators with a stroke of up to 1 millimeter [15]. As a result, the reticle chain can cover a large range with high positioning accuracy [25].

Compared to the reticle chain, the wafer chain depicted in Fig. 6 is heavier and moves in quarter speed of the reticle stage due to the lens magnification factor of 4. A dual-stage structure is adopted at this chain, as well. Allowing the wafer stage to cover a large range with high positioning accuracy [25]. It is noteworthy that the wafer stage has 6-DOFs and therefore is capable of realizing in-plane motion $\{x, y, \theta_z\}$ and out-of-plane motion $\{z, \theta_x, \theta_y\}$. The in-plane motion takes place during step-and scan portions of the lithography process, while the out-of-plane motion maintains focusing and leveling of the wafer during that process. Both the in-plane and the out-of-plane motions are coupled [2]. Coupling occurs via the plant dynamics and not via the control design that assumes a perfectly decoupled system. Therefore, a 6-DOFs analysis must be carried out.

In addition, the optics box – within the optics chain depicted in Fig. 7 – usually utilizes active optical elements such as motion controlled mirrors with suitable control and measurement systems to realize an adaptive optics system [24] that corrects for lens aberrations and setpoint-induced vibration. In this study, for the aim of simplicity, the optics elements of the optics box have been

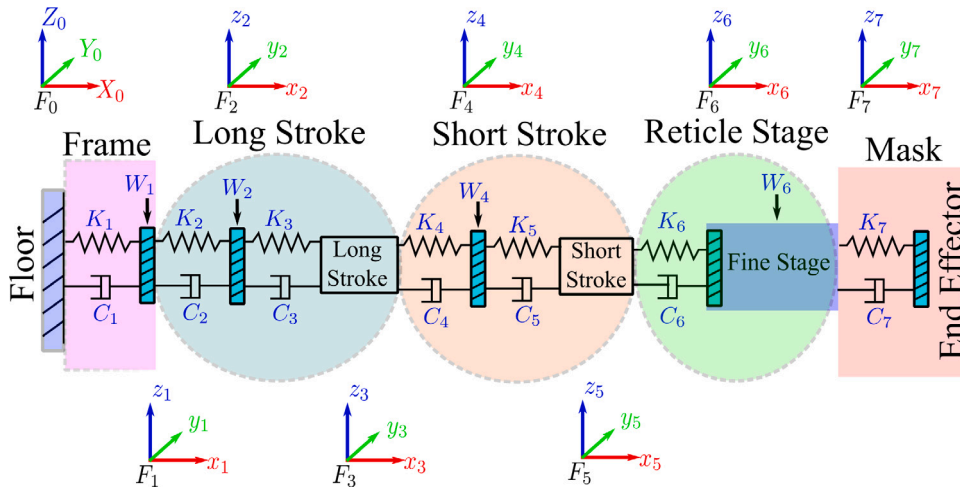


Fig. 5. A typical open-loop reticle chain lumped mass-spring-damper model and the proposed fine stage. Note that the chain index, i.e., $j = 1$ is omitted for clarity.

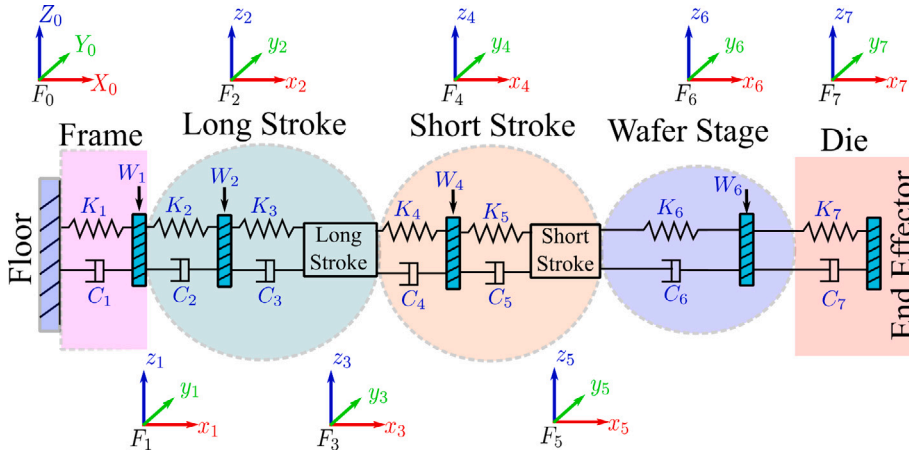


Fig. 6. A typical open-loop wafer chain lumped mass-spring-damper model. Note that the chain index, i.e., $j = 3$ is omitted for clarity.

approximated as a single lens element without any motion control. More complicated models can be obtained by considering each individual lens element with active motion control, this can be achieved by assigning a fixed-body frame at each lens [6], but this is out of our scope at the current time.

Remark 2.1. The connections between stages, i.e. joints surfaces, within a chain are realized using a low mass material that is modeled according to Kelvin-Voigt model as a parallel spring-damper system. Despite its simplicity, the Kelvin-Voigt model can be sufficiently used in the low-fidelity machine model presented herein. More representative nonlinear models can be used with medium and high levels of model fidelity.

The generalized coordinate that is used to describe the configuration of the stages is $q_{ij} = [f_{ij}^T, \theta_{ij}^T]^T \in \mathbb{R}^{6 \times 1}$, where f_{ij} and θ_{ij} are the linear and angular position vectors, respectively, of the i th stage within the j th chain. Based on Fig. 3, the inertial translational position vector of the F_{ij} frame expressed with respect to the inertial frame F_0 can be defined as [26]

$$f_{ij}^0 = f_{(i-1)j}^0 + f_{ij} \quad (2.1)$$

where $i \geq 2$, $j \in \{1, 2, 3\}$, $f_{1j}^0 \equiv f_{1j}$, $\forall j$. Taking the first and second time derivatives of (2.1) yields the inertial velocity and acceleration, respectively, of the i th stage, as expressed below [26]

$$\begin{aligned} \dot{f}_{ij}^0 &= \dot{f}_{(i-1)j}^0 + \dot{f}_{ij} + [\omega_{(i-1)j}] f_{ij} \\ \ddot{f}_{ij}^0 &= \ddot{f}_{(i-1)j}^0 + \ddot{f}_{ij} + 2[\omega_{(i-1)j}] \dot{f}_{ij} + B_{(i-1)j} f_{ij} \end{aligned} \quad (2.2)$$

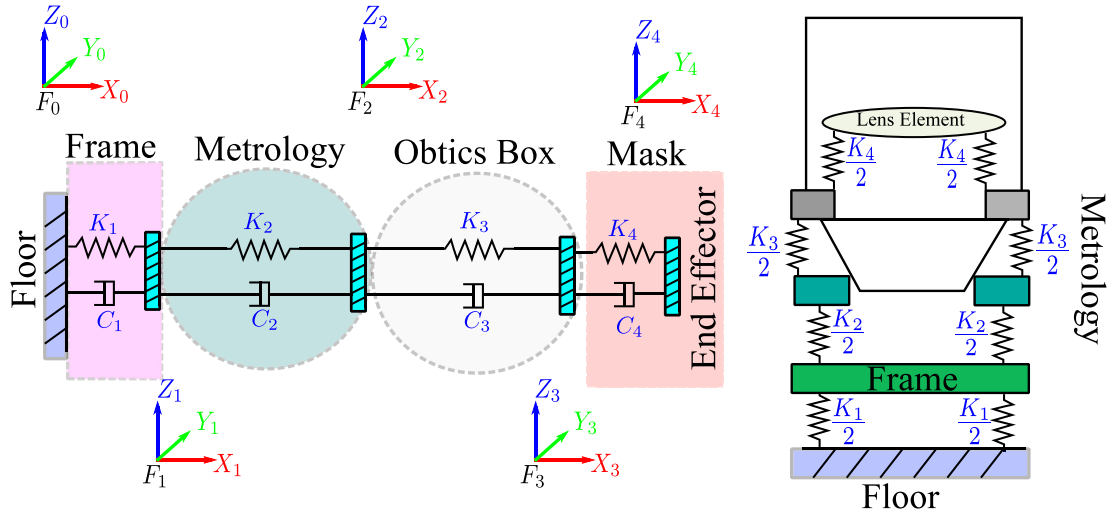


Fig. 7. A typical open-loop optics chain lumped mass-spring-damper model. Note that the chain index, i.e., $j = 2$ is omitted for clarity.

where ω_{ij} is the angular velocity of F_{ij} with respect to F_0 , $[\omega_{ij}]$ is its angular velocity given in the skew-symmetric representation, and B_{ij} is given by

$$B_{ij} = \begin{bmatrix} -\omega_y^2 - \omega_z^2 & -\dot{\omega}_z + \omega_x \omega_y & \dot{\omega}_y + \omega_x \omega_z \\ \dot{\omega}_z + \omega_y \omega_x & -\omega_x^2 - \omega_z^2 & -\dot{\omega}_x + \omega_y \omega_z \\ -\dot{\omega}_y + \omega_z \omega_x & \dot{\omega}_x + \omega_z \omega_y & -\omega_x^2 - \omega_y^2 \end{bmatrix}_{ij}$$

Using (2.1) and (2.2), the linear motion of the i th stage in the j th chain can be derived. Under the assumption of a stationary floor frame, the angular velocity and acceleration of the i th stage equal the sum of their corresponding quantities in the previous stages within the same chain plus those of the i th stage itself [26]. Hence, using Euler's equation, the equation of the angular motion of the i th stage in the j th chain expressed in F_0 frame is given $\forall i = 2, 3, \dots, N_j$ as

$${}^0I_{ij}\dot{\omega}_{ij} + ([\omega_{ij}]^0I_{ij})\omega_{ij} = m_{ij}^0 \quad (2.3)$$

where N_j is the number of stages in the j th-chain, and ${}^0I_{ij} \in \mathbb{R}^{3 \times 3}$ is the inertia matrix calculated at the CoG of F_{ij} and aligned with respect to F_0 [26]. According to Steiner's theorem [27], this inertia matrix can be calculated using the following algorithm

Algorithm 1: Mass and Inertia aggregation of the j^{th} open-loop chain with N_j -stages from the end-effector stage to the i^{th} stages

Result: ${}^0I_{ij}$

initialization ${}^0I_{N_j} = {}^0R_{N_j} {}^{N_j}\tilde{I}_{N_j} {}^0R_{N_j}^T$;

if $2 \leq i < N_j$ **then**

for $k = N_j \dots i + 1$ **do**

$$\tilde{m}_{(k-1)_j} = \sum_{b=k}^{N_j} m_{b_j};$$

$${}^0I_{(k-1)_j} = {}^0I_{k_j} + {}^0R_{(k-1)_j} {}^{(k-1)_j}\tilde{I}_{(k-1)_j} {}^0R_{(k-1)_j}^T - \tilde{m}_{(k-1)_j} [f_{k_j}]^2;$$

end

end

where ${}^0R_{ij} \in \mathbb{R}^{3 \times 3}$ is the non-singular rotation matrix from F_{ij} to F_0 , $\tilde{m}_{ij}$ is the total mass seen by the i th stage, m_{ij} is its mass and ${}^{ij}\tilde{I}_{ij}$ is its inertia matrix written in its local coordinate frame. The base-frame inertia matrix is given as

$${}^0I_1 = {}^0R_1 {}^1\tilde{I}_1 {}^0R_1^T + \sum_{j=1}^3 \left({}^0I_{2_j} - m_{2_j} [f_{2_j}]^2 \right) \quad (2.4)$$

2.1. Machine base-frame modeling

Let u_1 and $m_1 \in \mathbb{R}^{3 \times 1}$ represent the force and moment vectors, respectively, acting on the CoG of the machine base-frame. In addition, let its stiffness and damping matrices expressed in F_0 be given as K_{ij} and $C_{ij} \in \mathbb{R}^{3 \times 3}$, respectively. Therefore, the linear

motion of the base-frame can be characterized by

$$m_0 \ddot{f}_1 + {}^0\bar{C}_1 \dot{f}_1 + {}^0\bar{K}_1 f_1 - \sum_{j=1}^3 \left\{ {}^0\hat{C}_{2j} \dot{f}_{2j} + {}^0\hat{K}_{2j} f_{2j} \right\} = u_1^0 \quad (2.5)$$

where u_1^0 is the sum of all active forces acting on F_1 written with respect to F_0 , and

$$\begin{aligned} m_0 &= m_1 + \sum_{j=1}^3 \left(m_B^j + \sum_{i=2}^{N_j} m_{i,j} \right) \\ {}^0\bar{C}_1 &= {}^0C_1, {}^0\bar{K}_1 = {}^0K_1, {}^0\hat{C}_{2j} = {}^0C_{2j} \\ {}^0\hat{K}_{2j} &= {}^0K_{2j} + {}^0C_{2j} [\omega_1] \end{aligned}$$

where m_B^j denotes the balance mass, if any, involved in the j th chain. Note that the stiffness matrix $K_{i,j}$ must be expressed with respect to the floor frame F_0 , as [28] ${}^0K_{i,j} = {}^0R_{i,j} K_{i,j} {}^0R_{i,j}^T$, and the damping matrix $C_{i,j}$ as ${}^0C_{i,j} = {}^0R_{i,j} C_{i,j} {}^0R_{i,j}^T$.

2.2. The overall machine model

The linear motion of the 2nd stage in the j th chain expressed in F_0 frame is given as

$$\ddot{m}_{2j} \sum_{k=1}^2 \ddot{f}_{k,j} + {}^0\bar{C}_{2j} \dot{f}_{2j} + {}^0\bar{K}_{2j} f_{2j} - \left\{ {}^0\hat{C}_{3j} \dot{f}_{3j} + {}^0\hat{K}_{3j} f_{3j} \right\} = u_{2j}^0 \quad (2.6)$$

Similarly, the linear motion of the i th stage in the j th chain expressed in F_0 frame is given for $i = 3, \dots, N_j - 1$ as

$$\begin{aligned} \ddot{m}_{i,j} \sum_{k=1}^i \ddot{f}_{k,j} + {}^0\bar{C}_{i,j} \dot{f}_{i,j} + {}^0\bar{K}_{i,j} f_{i,j} + \dots \\ \sum_{k=2}^{i-1} \left\{ {}^0\hat{C}_{i,j,k} \dot{f}_{k,j} + {}^0\hat{K}_{i,j,k} f_{k,j} \right\} - \left\{ {}^0\hat{C}_{(i+1),j} \dot{f}_{(i+1),j} + {}^0\hat{K}_{(i+1),j} f_{(i+1),j} \right\} = u_{i,j}^0 \end{aligned} \quad (2.7)$$

and for $i = N_j$

$$\ddot{m}_{N_j} \sum_{k=1}^{N_j} \ddot{f}_{k,j} + {}^0\bar{C}_{N_j} \dot{f}_{N_j} + {}^0\bar{K}_{N_j} f_{N_j} + \sum_{k=1}^{N_j-1} \left\{ {}^0\hat{C}_{i,j,k} \dot{f}_{k,j} + {}^0\hat{K}_{i,j,k} f_{k,j} \right\} = 0 \quad (2.8)$$

where $\ddot{m}_{i,j}$, ${}^0\bar{C}_{i,j}$, ${}^0\bar{K}_{i,j}$, ${}^0\hat{C}_{i,j,k}$, ${}^0\hat{K}_{i,j,k}$, ${}^0\hat{C}_{(i+1),j}$, ${}^0\hat{K}_{(i+1),j}$, $m_{i,j}^0$, and $u_{i,j}^0$ are given in Appendix A which shows how the angular motion does affect these quantities. It is assumed here that the gravity effect is compensated by a suitable gravity compensator and therefore will not be considered in the modeling process. Using (2.3), the overall dynamics of the j th chain can be expressed as

$$\begin{bmatrix} M_j \ddot{p}_j + N_j \dot{p}_j + Q_j p_j \\ I_j \ddot{\Omega}_j + D_j \Omega_j \end{bmatrix} = \begin{bmatrix} \ddot{u}_j^0 \\ \ddot{m}_j^0 \end{bmatrix} \quad (2.9)$$

where

$p_j = [f_2^T, \dots, f_{N_j}^T]^T$, $\Omega_j = [\omega_2^T, \dots, \omega_{N_j}^T]^T$, $\ddot{u}_j^0 = [u_2^{0T}, \dots, u_{N_j}^{0T}]^T$, $\ddot{m}_j^0 = [m_2^{0T}, \dots, m_{N_j}^{0T}]^T$, $I_j = \text{diag} \{ {}^0I_2, \dots, {}^0I_{N_j} \}$, $D_j = \text{diag} \{ [\omega_2] {}^0I_2, \dots, [\omega_{N_j}] {}^0I_{N_j} \}$, and the mass matrix is given as

$$M_j = \begin{bmatrix} \ddot{m}_2 e_3 & \mathbf{0}_3 & \mathbf{0}_3 & \dots & \mathbf{0}_3 \\ \ddot{m}_3 e_3 & \ddot{m}_3 e_3 & \mathbf{0}_3 & \dots & \vdots \\ \vdots & \vdots & \ddots & \mathbf{0}_3 & \vdots \\ \ddot{m}_{N_j-1} e_3 & \dots & \dots & \ddot{m}_{N_j-1} e_3 & \mathbf{0}_3 \\ \ddot{m}_{N_j} e_3 & \dots & \dots & \dots & \ddot{m}_{N_j} e_3 \end{bmatrix}$$

with $e_k, \mathbf{0}_k \in \mathbb{R}^{k \times k}$ denoting the identity and zero matrices, and both N_j and Q_j share the same structure given as

$$H_j = \begin{bmatrix} {}^0\hat{H}_2 & -{}^0\hat{H}_{2,3} & \mathbf{0}_3 & \dots & \mathbf{0}_3 \\ {}^0\hat{H}_{3,2} & {}^0\hat{H}_3 & -{}^0\hat{H}_{3,4} & \mathbf{0}_3 & \vdots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ {}^0\hat{H}_{N_j-1,2} & {}^0\hat{H}_{N_j-1,3} & \dots & {}^0\hat{H}_{N_j-1} & -{}^0\hat{H}_{N_j-1,N_j} \\ {}^0\hat{H}_{N_j,2} & {}^0\hat{H}_{N_j,3} & \dots & {}^0\hat{H}_{N_j,N_j-1} & {}^0\hat{H}_{N_j} \end{bmatrix}$$

where $H \in \{N, Q\}$ in which ${}^0\hat{H}_i, {}^0\hat{H}_{i,j}$ can be found using (2.6) through (2.8). In (2.9), $N_2 = 4$ and $N_1 = N_3 = 7$ are used. Thus, using (2.5), the overall machine linear motion can be described by

$$M \ddot{P} + N \dot{P} + Q P = u \quad (2.10)$$

with $P = [f_1^T, p_1^T, p_2^T, p_3^T]^T$, $u = [u_1^0, \bar{u}_1^0, \bar{u}_2^0, \bar{u}_3^0]^T$ and

$$M = \begin{bmatrix} m_0 e_3 & 0_{3 \times 18} & 0_{3 \times 9} & 0_{3 \times 18} \\ M_{m,1} & M_1 & 0_{18 \times 9} & 0_{18 \times 18} \\ M_{m,2} & 0_{9 \times 18} & M_2 & 0_{9 \times 18} \\ M_{m,3} & 0_{18 \times 18} & 0_{18 \times 9} & M_3 \end{bmatrix}, H = \begin{bmatrix} {}^0 \hat{H}_1 & H_{m,1} & H_{m,2} & H_{m,3} \\ 0_{18 \times 3} & H_1 & 0_{18 \times 9} & 0_{18 \times 18} \\ 0_{9 \times 3} & 0_{9 \times 18} & H_2 & 0_{9 \times 18} \\ 0_{18 \times 3} & 0_{18 \times 18} & 0_{18 \times 9} & H_3 \end{bmatrix}$$

where $M_{m,j} = [\bar{m}_2 e_3, \dots, \bar{m}_{N_j} e_3]^T$, $H_{m,2} = [{}^0 \hat{H}_2, 0_3, 0_3]$, $H_{m,j \in \{1,3\}} = [{}^0 \hat{H}_{2_j}, 0_3, 0_3, 0_3, 0_3, 0_3]$ and N_2 , Q_2 , and M_2 can be found from (2.9) with $N = 4$. Writing (2.10) in a state-space form with state vector $\mathbb{P} = [P^T, \dot{P}^T]^T$, yields

$$\begin{bmatrix} \dot{P} \\ \ddot{P} \end{bmatrix} = \begin{bmatrix} 0_{48} & e_{48} \\ M^{-1}Q & M^{-1}N \end{bmatrix} \begin{bmatrix} P \\ \dot{P} \end{bmatrix} + \begin{bmatrix} 0_{48} \\ M^{-1} \end{bmatrix} u \quad (2.11)$$

$$\dot{\mathbb{P}} = A\mathbb{P} + Bu$$

Similarly, the machine rotational motion can be written using $\Omega = [\omega_1^T, \Omega_1^T, \Omega_2^T, \Omega_3^T]^T$ as

$$\begin{bmatrix} {}^0 I_1 & 0_{3 \times 18} & 0_{3 \times 9} & 0_{3 \times 18} \\ 0_{18 \times 3} & I_1 & 0_{18 \times 9} & 0_{18 \times 18} \\ 0_{9 \times 3} & 0_{9 \times 18} & I_2 & 0_{9 \times 18} \\ 0_{18 \times 3} & 0_{18 \times 18} & 0_{18 \times 9} & I_3 \end{bmatrix} \begin{bmatrix} \dot{\omega}_1 \\ \dot{\Omega}_1 \\ \dot{\Omega}_2 \\ \dot{\Omega}_3 \end{bmatrix} + \begin{bmatrix} {}^0 D_1 & 0_{3 \times 18} & 0_{3 \times 9} & 0_{3 \times 18} \\ 0_{18 \times 3} & D_1 & 0_{18 \times 9} & 0_{18 \times 18} \\ 0_{9 \times 3} & 0_{9 \times 18} & D_2 & 0_{9 \times 18} \\ 0_{18 \times 3} & 0_{18 \times 18} & 0_{18 \times 9} & D_3 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \Omega_1 \\ \Omega_2 \\ \Omega_3 \end{bmatrix} = \begin{bmatrix} m^0 \\ \bar{m}^0 \\ \bar{m}^0 \\ \bar{m}^0 \end{bmatrix} \quad (2.12)$$

$$I\dot{\Omega} + D\Omega = m$$

where ${}^0 D_1 = [\omega_1]^0 I_1$. Recalling (2.5) through (2.8), (2.11) can be written as

$$\dot{\mathbb{P}} = A(\Omega, \dot{\Omega})\mathbb{P} + Bu = A_0\mathbb{P} + Bu + A_1(\Omega, \dot{\Omega})\mathbb{P} \quad (2.13)$$

which represents a linear parameter-varying system with A_0 holds the nominal terms, i.e. when $\Omega = \dot{\Omega} = 0$, and A_1 holds the varying parameters, i.e. when $\Omega, \dot{\Omega} \neq 0$.

As depicted in Figs. 1 and 2, the metrology frame represents the measurement reference frame in the lithography machine, and it is connected to the machine base using air mounts in order to be isolated from the base vibrations [4]. Mathematically, designing the sensor matrix (C), the machine states of interest ($\mathbb{Y}$) are given by $\mathbb{Y} = C\mathbb{P}$. Therefore, (2.13) is rewritten as

$$\begin{aligned} \dot{\mathbb{P}} &= A_0\mathbb{P} + Bu + A_1(\Omega, \dot{\Omega})\mathbb{P} \\ \mathbb{Y} &= C\mathbb{P} \end{aligned} \quad (2.14)$$

Now, characterization of the machine's controllability, observability and various stability, performance and control aspects [29] can be studied based on (2.14). To simulate the machine, both (2.12) and (2.14) are used.

It is noteworthy that the distribution of actuators determines their abilities to generate moments. In a lithography machine, the long-stroke is still often realized under the H-configuration where three linear actuators move the short-stroke stage in the x - y plane [2]. The voice coil motors or Lorentz short-stroke actuators achieve 6-DOF motion when they are distributed correctly. The layouts outlined in [2] are used in this study.

Remark 2.2. Different from regular motion systems [30], in high-precision motion systems including lithography machines, air bearings are widely used to provide contact-free motion with zero friction [31] when the airflow friction between the bearing surfaces is deemed negligible. Ignoring the airflow friction, both sliding and rolling frictional resistance are discarded in the presented low-fidelity machine model.

2.3. Numeric values of parameters

When modeling a specific lithography machine, input-output data is collected from identification experiments and represented either in time or frequency domains. Then, multiple-input multiple-output (MIMO) parametric identification techniques, see for example [32–35], can be applied to come up with the numeric values of the model parameters. For Estimating state-space models with structured parameterization – like (2.14) especially when $A_1 = 0$ –, some software packages provide dedicated solutions, see for example the MATLAB function `idss`. In this study, parametric identification is beyond the current scope, and therefore numeric values of a hypothetical lithography machine parameters are provided in Appendix B.

3. Fine stage as a positioning-error regulator

Due to various causes, positioning errors take place in forms of tracking, alignment, synchronization and in-plane or out-of-plane errors. Several efforts were reported to reduce such errors by, for example, on-line and repetitive tuning of the positioning controllers [36], redesigning the reference profiles, adding feedforward controllers of higher-order time derivatives [23]; however, other solutions are still being sought.

As a second contribution, this study proposes a novel idea of introducing additional fine stages to enhance lithography process, as depicted in Fig. 8. Therefore, a generalized fine stage model must be derived and then combined with the developed machine model; so that the needed design requirements of any customized fine stage can be obtained.

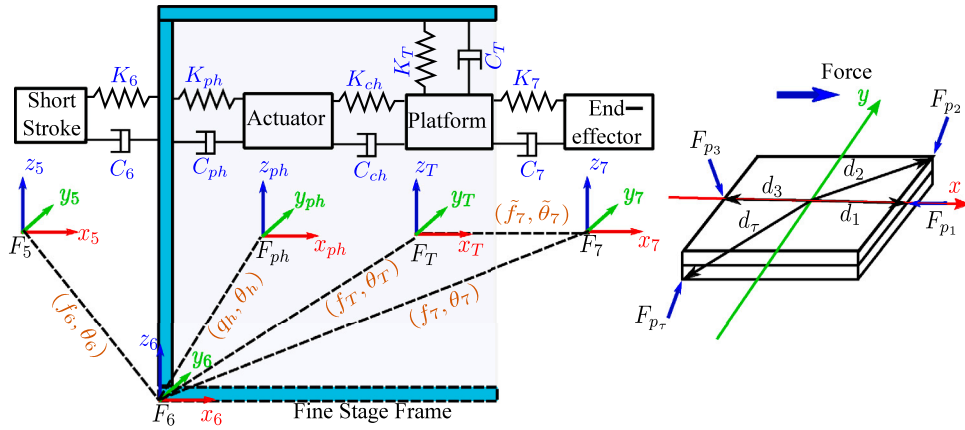


Fig. 8. To the left: An equivalent lumped mass–spring–damper model of a generic fine stage attached to the short-stroke of the desired chain. The chain index is omitted where it can be $j = 1, 3$ for the reticle and wafer chains, respectively. To the right, the free-body diagram of the platform that holds the chain end-effector.

3.1. Dynamics model of the fine stage

The motion of the fine stage can be modeled using linear lumped mass–spring–damper systems as depicted in Fig. 8. Let the fine stage coordinates be given as $\mathbf{f} = [q_1^T, \dots, q_\tau^T, \mathbf{f}_T^T, \boldsymbol{\theta}_T^T]^T$ where $q_i \in \mathbb{R}^{3 \times 1}$ is the displacement of the i th actuator with $i = 1, 2, \dots, \tau$, $\tau \geq 1$ is the number of the actuators in the fine stage, and $\mathbf{f}_T = [f_{Tx}, f_{Ty}, f_{Tz}]^T \in \mathbb{R}^{3 \times 1}$ and $\boldsymbol{\theta}_T = [\theta_{Tx}, \theta_{Ty}, \theta_{Tz}]^T \in \mathbb{R}^{3 \times 1}$ are the linear and angular displacements of the platform, respectively. Let ${}^0F_{pi} \in \mathbb{R}^{3 \times 1}$ be the force generated by the i th actuator expressed in F_0 frame, see Fig. 8. In this section, we will drop the chain index for simplicity.

Remark 3.1. Based on the model fidelity level sought and the assumed range of operation, associated nano-positioner nonlinearities can be included or excluded. In this study, we only include the linear range of operation of the involved actuators based on the developed low-fidelity model of the lithography machine.

The linear motion of the fine stage frame can be given as

$$\begin{aligned} \bar{m}_6 \sum_{j=1}^5 \{ \ddot{\mathbf{f}}_j + 2[\boldsymbol{\omega}_{j-1}] \dot{\mathbf{f}}_j + [\mathbf{B}_{j-1}] \mathbf{f}_j \} + \bar{m}_6 \ddot{\mathbf{f}}_6 + \{ {}^0C_6 + 2\bar{m}_6[\boldsymbol{\omega}_5] \} \dot{\mathbf{f}}_6 + \{ {}^0K_6 + {}^0C_6[\boldsymbol{\omega}_5] + \dots \\ \bar{m}_6[\mathbf{B}_5] \} \mathbf{f}_6 + \sum_{h=1}^{\tau} \{ {}^0C_{ph} \dot{\mathbf{q}}_h + ({}^0K_{ph} + {}^0C_{ph}[\boldsymbol{\omega}_6]) \mathbf{q}_h \} - {}^0C_T \dot{\mathbf{f}}_T - ({}^0K_T + {}^0C_T[\boldsymbol{\omega}_6]) \mathbf{f}_T = \mathbf{0}_{3 \times 1} \end{aligned} \quad (3.1)$$

where $\bar{m}_6 = m_6 + m_{p\tau} + \bar{m}_T$ is the total mass of the fine stage with $m_{p\tau}$ as the total mass of the identical actuators, and $\bar{m}_T$ is the total mass of the platform including the end-effector mass. The rotational motion of the fine stage frame can be given as

$$\tilde{I}_6 \ddot{\boldsymbol{\theta}}_6 + {}^0C_{\theta_6} \dot{\boldsymbol{\theta}}_6 + {}^0K_{\theta_6} \boldsymbol{\theta}_6 = \mathbf{0}_{3 \times 1} \quad (3.2)$$

where ${}^0C_{\theta_6}$ and ${}^0K_{\theta_6}$ are the rotational damping and stiffness matrices, respectively, of the fine stage frame. In addition, the linear motion of the h th actuator can be given as

$$\begin{aligned} m_p \sum_{j=1}^6 \{ \ddot{\mathbf{f}}_j + 2[\boldsymbol{\omega}_{j-1}] \dot{\mathbf{f}}_j + [\mathbf{B}_{j-1}] \mathbf{f}_j \} + m_p \ddot{\mathbf{q}}_h + \{ {}^0C_{ch} + {}^0C_{ph} + 2m_p[\boldsymbol{\omega}_6] \} \dot{\mathbf{q}}_h \\ + \{ {}^0K_{ch} + {}^0K_{ph} + ({}^0C_{ch} + {}^0C_{ph})[\boldsymbol{\omega}_6] + m_p[\mathbf{B}_6] \} \mathbf{q}_h - {}^0C_{ch} \dot{\mathbf{f}}_T - ({}^0K_{ch} + {}^0C_{ch}[\boldsymbol{\omega}_6]) \mathbf{f}_T = {}^0F_{ph} \end{aligned} \quad (3.3)$$

where all actuators have the same mass m_p . Moreover, the rotational motion of the h th actuator is similar to (3.2).

The linear motion of the fine stage platform can be given as

$$\begin{aligned} \bar{m}_T \sum_{j=1}^6 \{ \ddot{\mathbf{f}}_j + 2[\boldsymbol{\omega}_{j-1}] \dot{\mathbf{f}}_j + [\mathbf{B}_{j-1}] \mathbf{f}_j \} + \bar{m}_T \ddot{\mathbf{f}}_T + \left\{ {}^0C_7 + {}^0C_T + \sum_{h=1}^{\tau} {}^0C_{ch} \right\} \dot{\mathbf{f}}_T \\ + \left\{ {}^0K_7 + {}^0K_T + ({}^0C_7 + {}^0C_T)[\boldsymbol{\omega}_6] + \sum_{h=1}^{\tau} \{ {}^0K_{ch} + {}^0C_{ch}[\boldsymbol{\omega}_6] \} \right\} \mathbf{f}_T \\ - \sum_{h=1}^{\tau} \{ {}^0K_{ch} + {}^0C_{ch}[\boldsymbol{\omega}_6] \} \mathbf{q}_h - \sum_{h=1}^{\tau} {}^0C_{ch} \dot{\mathbf{q}}_h - ({}^0K_7 + {}^0C_7[\boldsymbol{\omega}_6]) \mathbf{f}_7 = \mathbf{0}_{3 \times 1} \end{aligned} \quad (3.4)$$

and its rotational motion is given as

$$\tilde{I}_T \ddot{\theta}_T + {}^0C_{\theta_T} \dot{\theta}_T + {}^0K_{\theta_T} \theta_T = \sum_{h=1}^{\tau} [d_h] {}^0\tilde{F}_{p_h} \quad (3.5)$$

where ${}^0C_{\theta_T}$ and ${}^0K_{\theta_T}$ are the rotational damping and stiffness matrices of the platform, d_h is the distance between the h th actuator and the CoG of the platform, and $\tilde{m}_T = m_T + m_7$ is its total mass with m_7 as the mass of the end-effector and ${}^0\tilde{F}_{p_h}$ is the h th actuator filtered force. The force ${}^0\tilde{F}_{p_h}$ along the x -direction is given in the frequency domain as

$$\frac{f_{T_x}(s)}{F_{p_{hx}}(s)} = \frac{K_c + C_c s}{(m_p m_T) s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0} \quad (3.6)$$

where $a_4 = m_p + m_T$, $a_3 = a_4 C_c + m_T (C_p + C_c)$, $a_2 = a_4 K_c + C_p C_c + m_T (K_p + K_c)$, $a_1 = K_c C_p + k_p C_c$, $a_0 = K_p K_c$ and the other components of ${}^0\tilde{F}_{p_h}$ follow similarly. Note that at steady-state, the relation between f_{T_x} and $F_{p_{hx}}$ is given by

$$\lim_{s \rightarrow 0} \frac{f_{T_x}(s)}{F_{p_{hx}}(s)} = \frac{1}{K_p} \quad (3.7)$$

Using (3.6) and (3.7), input maps can be used to build the closed-loop control schematic, i.e compliance rather than mass behavior at low frequencies. Therefore, the fine stage can regulate the leftover tracking errors by utilizing the fine stage platform position and orientation information. Similarly, the linear motion of the end-effector can be given as

$$m_7 \sum_{j=1}^6 \{ \ddot{f}_j + 2 [\omega_{j-1}] \dot{f}_j + [B_{j-1}] f_j \} + m_7 \ddot{f}_7 + \{ {}^0C_7 + 2m_7 [\omega_6] \} \dot{f}_7 + \{ {}^0K_7 + {}^0C_7 [\omega_6] + m_7 [B_6] \} f_7 - {}^0C_7 \dot{f}_T - \{ {}^0K_7 + {}^0C_7 [\omega_6] \} f_T = \mathbf{0}_{3 \times 1} \quad (3.8)$$

and its rotational motion is similar to (3.2) with appropriate parameters.

3.2. The machine and fine stage combined model

To obtain the dynamics of the j th chain after including the fine stage, let the new states vector of the j th chain given by $p_j = [f_2^T, f_3^T, f_4^T, f_5^T, f_6^T, q_1^T, \dots, q_r^T, f_T^T, f_7^T]^T$ where $j = 1, 3$ for the reticle and wafer chains, respectively. Consequently, using (3.1) through (3.8), the j th chain dynamics – given by (2.9) – and the overall machine dynamics – given by (2.14) – can be reconstructed accordingly.

Remark 3.2. At this point, it is possible to select the desired chain to embed the fine stage. For example, the fine stage can be introduced to the reticle chain as depicted in Fig. 5, to the wafer chain depicted in Fig. 6, or more interestingly to both of them. The last option can be made such that the mass, the working space and the complexity of an individual fine stage can be reduced significantly. For example, let the fine stage at the reticle chain moves in the $\{x, y, \theta_z\}$ -directions, while the fine stage at the wafer chain moves in $\{z, \theta_x, \theta_y\}$ -directions. Therefore, both fine stages combined can compensate for the positioning errors in 6-DOF with considerably lower complexity once compared to a single 6-DOF fine stage residing at a particular chain.

Recalling Remark 3.2, we may consider introducing two fine stages; a fine stage at the reticle chain that moves in the $\{x, y, \theta_z\}$ -directions, and another fine stage at the wafer chain that moves in $\{z, \theta_x, \theta_y\}$ -directions. These stages are depicted in Fig. 9 which shows the resultant forces, torques, and the used actuators layout. Being excessively over-actuated, see for example [14], the motion of the reticle chain fine stage can be made almost decoupled; i.e. less than 15% coupling, in the $\{x, y, \theta_z\}$ -directions. The realization of the fine stages using the proposed machine and fine stage models will be covered in a separate study.

4. Interaction among chains

Since the operating principle of a lithography machine requires the interaction between the reticle, lens and wafer stages during the scanning motion, it is imperative to find a mathematical model for such an interaction. This model will help in better understanding the propagation of errors that mostly affect the machine throughput as well as fading and imaging errors. This can be done by tracing the light ray that passes through the reticle to the wafer through the lens.

4.1. Ray-trace modeling

Fig. 10 shows the ray tracing for the ideal case in which the three chains are horizontally aligned with respect to each other and no disturbances are active. Approximating the optical system as a thin positive lens element with negligible rotational motion, it is possible to use the Gaussian Lens Formula [24] to derive the relation between the chains end-effectors in the z -direction as

$$\frac{1}{f_0} = \frac{1}{h_1} + \frac{1}{h_2} = \frac{(f_{6_1}^0 + f_{7_1} - f_{6_3}^0 - f_{7_3})_z}{(f_{6_1}^0 + f_{7_1} - f_{4_2}^0)_z (f_{4_2}^0 - f_{6_3}^0 - f_{7_3})_z} \quad (4.1)$$

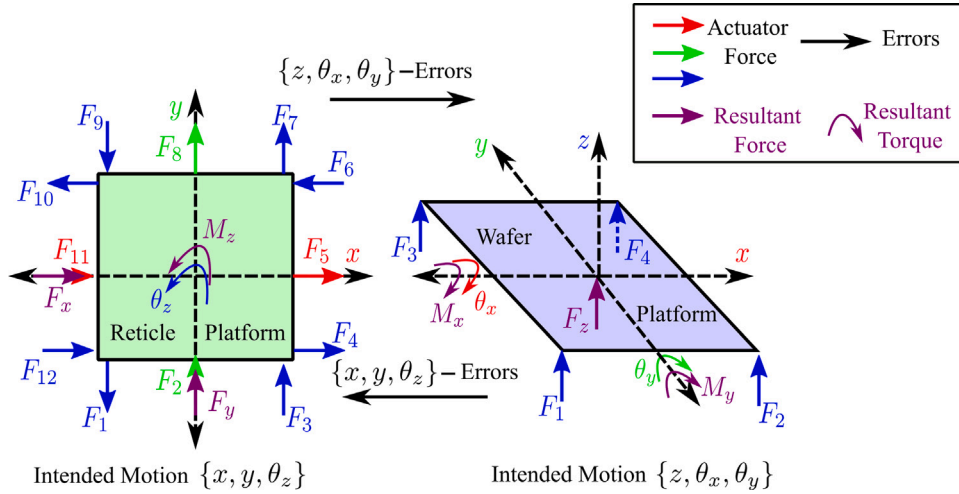


Fig. 9. Two fine stages proposed at the reticle chain (left) and at the wafer chain (right) with exchanged error signal among them. The fine stage at the reticle chain utilizes twelve in-plane actuators, while the fine stage at the wafer chain utilizes only four actuators in the z -direction.

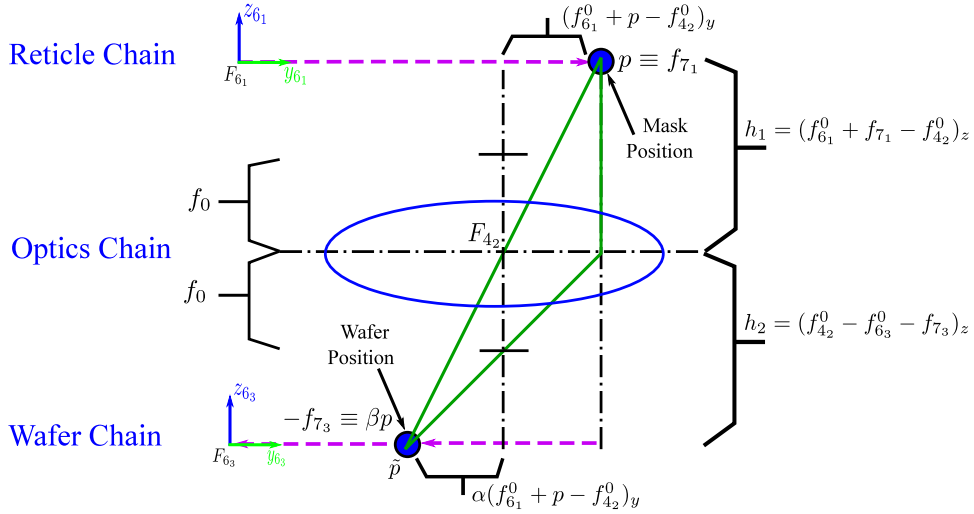


Fig. 10. Chains interaction when, for simplicity, the end-effectors are horizontally aligned and the wafer is in the lens element focal plane. The optics are approximated as a positive thin lens element with a constant focal point.

where f_0 is the focal point of the equivalent lens. From the similarity of triangles, we have $\alpha h_1 = h_2$. Then we may write

$$\alpha(f_{61}^0 + f_{71} - f_{42}^0)_z + (f_{42}^0 - f_{63}^0 - f_{73})_z = 2(\alpha + 1)f_0 \quad (4.2)$$

where $\alpha > 0$, specifically 0.25, is the value of the fixed lens demagnification given as [24] $\alpha = -h_2/h_1$ where the negative sign means that the image is in the opposite side of the lens compared to the pattern or object projected. To have a real image, we must have $h_1 > f_0$ [24].

Compared to the horizontal positioning error, the focus error can be allowed to be 103 times larger [37]. Besides positioning errors in the z -direction, wafer unflatness can affect the depth of focus measures of the lithography process [38]. This problem can be solved by adjusting the focal point of the optical system using adaptive optics [24], moving the reticle stage, the wafer stage or both of them vertically such that (4.1) is satisfied, or by bending the photomask [38]. The focus error effect will be handled in a separate study.

4.2. Fine stages and chains interactions

To derive the chains interaction kinematical relation along the y -direction, from Fig. 10, we may write

$$f_{63,y}^0 - f_{42,y}^0 = \beta p_y + \alpha(f_{61,y}^0 + p_y - f_{42,y}^0) \quad (4.3)$$

where $\beta > 0$ is a uniform scaling factor. Simplifying (4.3) yields the y -component of the *modified synchronization error* ($\tilde{e}_{syn}$) surface associated with the equation given as

$$(\alpha f_{61,y}^0 - f_{63,y}^0) + (1 - \alpha)f_{42,y}^0 - f_{73,y} + \alpha f_{71,y} = 0 \quad (4.4)$$

Note that when the motions of the lens element, the mask and the wafer substrate are ignored, then the synchronization error (e_{syn}) usually used, cf. [39], will be associated with the equation given as

$$\alpha f_{61,y}^0 - f_{63,y}^0 = 0 \quad (4.5)$$

Using (2.1), we can rewrite (4.4) as

$$\alpha(f_{61,y}^0 - f_{42,y}^0) + (f_{42,y}^0 - f_{63,y}^0) - f_{73,y} + \alpha f_{71,y} = \alpha(f_{71,y}^0 - f_{42,y}^0) + (f_{42,y}^0 - f_{73,y}^0) = 0 \quad (4.6)$$

which helps in capturing the chains interaction with respect to the lens coordinates. If (4.6) does not hold, then undesired errors will build up. One way to mitigate such error is through the synchronization of the open-chains. This synchronization implies that the machine needs to *virtually* operate as a closed-chain parallel manipulator robotic system, see [40,41]. Since all the chains share the same root, namely the frame, its motion will not affect the chains interaction and consequently the exposure process when the mask and the wafer motions are ignored. Therefore, less stringent conditions on anti-vibration systems can take place. Moreover, the lens and – by extension – the optics box need not be stationary anymore. These observations may lead to better machine designs where structural rigidity, vibration and mass dilemmas [24] can be addressed differently resulting in reduced machine costs and increased throughput. If the mask and the wafer motions are available and to be included as in (4.6), then the effect of the machine frame motion can be regulated using the proposed smart-board. At this point, it is useful to use the absolute positions of the mask, lens and wafer. The kinematical relation of the chains interaction along the x -direction can be found similarly.

Remark 4.1. In this paper, instead of realizing the chains synchronization through cooperative control signals design, we will utilize the proposed fine stages. Synchronization through these stages allows the stages controllers to be designed independently with possibly minimum modification.

4.3. Fine stages reference signals

Recalling Fig. 8 and using both (4.2) and (4.6), the reference signal of a fine stage at the reticle chain can be derived from the modified synchronization error $\tilde{e}_{syn}$ as

$$f_{T1,\zeta} = \frac{1}{\alpha}((\alpha - 1)f_{42,\zeta}^0 + f_{73,\zeta}^0) - (f_{61,\zeta}^0 + \tilde{f}_{71,\zeta}) \quad (4.7)$$

where $\zeta \in \{x, y\}$ and $\tilde{f}_{71}$ is the mask position relative to the reticle fine stage platform and it is assumed to be available. When using the synchronization error e_{syn} , then

$$f_{T1,\zeta} = e_{syn} = \alpha f_{61,\zeta}^0 - f_{63,\zeta}^0 \quad (4.8)$$

The reticle fine stage reference signal in the θ_z -direction is given as

$$\theta_{T1,z} = \theta_{73,z} - \tilde{\theta}_{71,z} \quad (4.9)$$

where $\tilde{\theta}_{71}$ is the mask orientation relative to the reticle fine stage platform and it is assumed to be available. Considering the fine stages depicted in Fig. 9, the interaction between the fine stages can be captured by rewriting (4.7) and (4.9) as

$$f_{T1,\zeta} = \frac{1}{\alpha}((\alpha - 1)f_{42,\zeta}^0 + f_{63,\zeta}^0 + f_{T3,\zeta} + \tilde{f}_{73,\zeta}) - (f_{61,\zeta}^0 + \tilde{f}_{71,\zeta}) \quad (4.10)$$

$$\theta_{T1,z} = \theta_{T3,z} + \tilde{\theta}_{73,z} - \tilde{\theta}_{71,z} \quad (4.11)$$

For the wafer chain fine stage, the reference signals can be given as

$$f_{T3,z} = (1 - \alpha)f_{42,z}^0 + \alpha(f_{61,z}^0 + f_{T1,z} + \tilde{f}_{71,z}) - (f_{63,z}^0 + \tilde{f}_{73,z}) - 2(\alpha + 1)f_0 + z_l \quad (4.12)$$

$$\theta_{T3,\zeta} = \theta_{T1,\zeta} + \tilde{\theta}_{71,\zeta} - \tilde{\theta}_{73,\zeta} \quad (4.13)$$

where z_l is the wafer substrate level at the current die to be exposed.

5. Errors in lithography machines

Both overlay and contrast measures are important specifications in lithography machines. Overlay is defined as the ability of the lithography machine to expose two images on top of each other [2]. Overlay correlates with the lower-frequency part of the positioning errors of the stages during exposure, and it can be represented using moving average MA [2] filtering. Contrast can be partly characterized by moving standard deviation MSD filtering resulting in the higher-frequency part of the positioning errors during exposure [2].

For 38-nm half-pitch lithography, typically the $MA \leq 1$ nm, and the $MSD \leq 7$ nm [2] though it must be understood that several measures for MA and MSD are being used, typically expressed by different norms (e.g. infinity-norm) or different assumptions on the slit shape (box or arc-shaped). Since the reticle stage moves four times faster than the wafer stage, using the same profile (P_d) of point P shown in Fig. 3 during scanning, MA – and similarly MSD – can be used to assess the tracking errors of the reticle (e_r) and the wafer (e_w) stages during a scanning period in the y -direction as

$$\begin{aligned} MA_{r,y} &= \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} e_{r,y}(t) dt = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} (4P_{d,y} - f_{61,y}^0) dt \\ MA_{w,y} &= \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} e_{w,y}(t) dt = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} (P_{d,y} - f_{63,y}^0) dt \end{aligned} \quad (5.1)$$

with T represents the exposure time of typically 1.7187510×10^{-3} s for a slit length of 5.5×10^{-3} m and a reticle scanning velocity of 3.2 m/s. Using (4.8), the synchronization error is given as [39]

$$e_{syn} = e_{w,y} - 0.25e_{r,y} = 0.25f_{61,y}^0 - f_{63,y}^0 \quad (5.2)$$

and its associated MA and MSD are given as

$$MA_{syn} = MA_{w,y} - 0.25MA_{r,y} \quad (5.3)$$

$$MSD_{syn} = \sqrt{\frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} (e_{syn}(t) - MA_{syn})^2 dt} \quad (5.4)$$

In this study, we *desire* that $MA_{syn} \in [-1.25, 2]$ nm *during the scanning phase*. Note that to meet certain specifications, both MA_{syn} and MSD_{syn} can be optimized as outlined in [39].

6. Simulation results

In this section, *mainly* the synchronization error due to chains interaction will be evaluated for two identical hypothetical lithography machines where one of them has the fine stage(s) as novelty for the reticle and/or wafer stages. The frequency allocation of this machine without the fine stages is shown in Fig. 11 and the other needed numeric values are listed in Appendix B, and the machine is operating under proportional–integral–derivative (PID) controllers. The desired operating trajectories are basically the step and scan trajectories designed according to [2], and are depicted in Fig. 12. Corresponding fine stages of both the reticle and wafer stages have the same total mass, and both machines are operating under the same PID controllers and operating trajectories. Both (5.3) and (5.4) were minimized to tune the involved PID machine controllers.

Remark 6.1. In this study, PID-based feedback control was used. This should not limit the use of other control schemes usually investigated in the literature like feedforward [42] or iterative learning control [43], for example.

We will handle two cases, the first case studies the response of a machine that has only one fine stage introduced at the reticle chain and compare its performance to that of the machine without any fine stages. The second case deals with a machine with the two fine stages depicted in Fig. 9. Opposite to the wafer chain fine stage, the reticle chain fine stage has a controlled motion in the $\{x, y, \theta_z\}$ -directions, while the motion in the $\{z, \theta_x, \theta_y\}$ -directions is minimized by design but not actively controlled. Also, the fine stage at the reticle chain has twelve actuators, i.e. $\tau = 12$, while that at the wafer chain has only four actuators, i.e. $\tau = 4$. During the simulation, both (5.3) and (5.4) were used to evaluate the performance indices.

6.1. Case 1

Remark 6.2. To facilitate evaluating the proposed fine stage, we used two identical machines except at the reticle stage of the reticle chain. Compared to the – passive – reticle stage, the fine stage acts as an active positioning stage with increased stiffness and higher resonance frequencies due to the used actuators and the compliant mechanism design, in general. Both stages have the same mass and both machines are operated under identical control schemes.

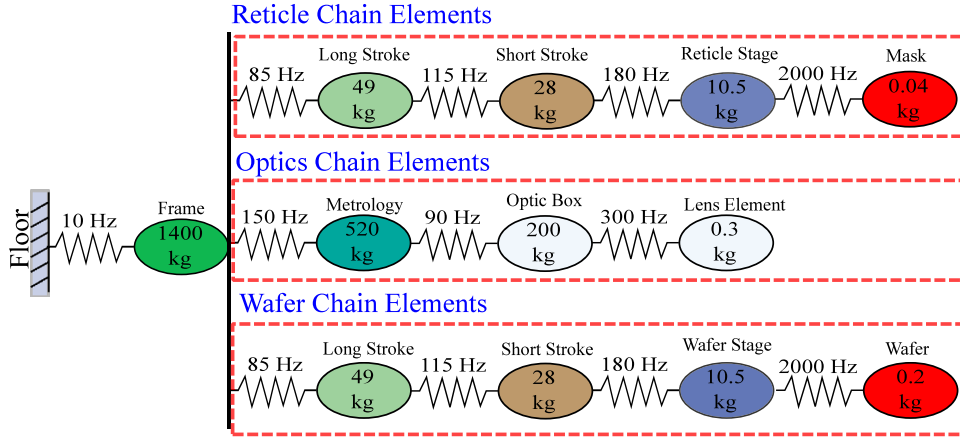


Fig. 11. Frequency allocation of the simulated lithography machine without the fine stages where the other needed numeric values are listed in [Appendix B](#).

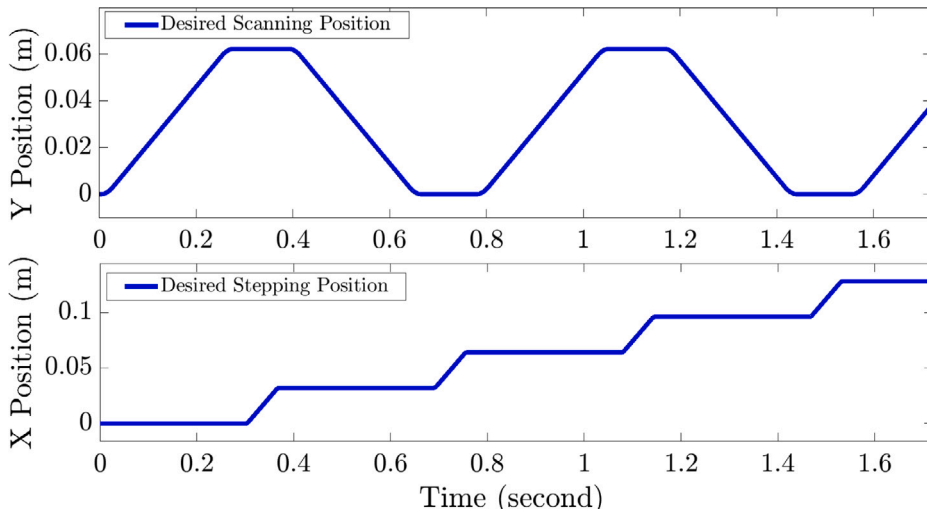


Fig. 12. The desired step-and-scan trajectories, relative to the lens element coordinates, used in the simulation as inspired by [\[2\]](#).

Recalling [Remark 6.1](#), the performance of the machine without fine stage can be enhanced by using feedforward in combination with feedback control. This may significantly reduce the tracking errors e_w and e_r . However, recalling [Remark 6.2](#), the results reported here are still valid since we assume any used control scheme to be identical in both machines and expect fine stage even has access to possibly higher bandwidth values than the passive stage (because of the higher resonance and stiffer connections). [Fig. 13](#) illustrates the block diagram of the closed-loop position control of the reticle chain including the proposed fine stage in the y -direction utilizing the tracking error signals.

The wafer trajectories during scanning and stepping are depicted in [Fig. 12](#). During scanning, both the reticle and wafer stages move in the y -direction (though in opposite direction) where the reticle stage speed is four times, i.e., $\alpha = 0.25$, larger than the wafer stage.

[Fig. 14](#) shows the evaluated performance indices of the two machines during the lithography process of four $32 \times 32 \text{ mm}^2$ dies. Interestingly, by using the synchronization error in [\(5.2\)](#) as a fine stage reference instead of only the reticle tracking error e_r given in [\(5.1\)](#), less-oscillatory and enhanced performance under the same fine stage controller parameters can be obtained as depicted in [Fig. 14a](#). [Fig. 15](#) shows the performance indices when using the modified synchronization error given in [\(4.6\)](#) mainly in the y -direction compared to the synchronization error given by [\(4.5\)](#). Clearly, using $\tilde{e}_{syn}$ offers improved performance compared to e_{syn} due to the additional information used. The lens element induced vibration is negligible because of the small energy propagation through the reticle and the machine frame that have huge masses compared to the reticle and wafer chains, see [Fig. 11](#). As depicted in [Fig. 16](#), reducing the stepping time results in an increased machine throughput while maintaining the performance. Optimizing the desired step-and-scan trajectories under the action of the fine stage will be a subject of future work. It is clear that using [\(4.6\)](#),

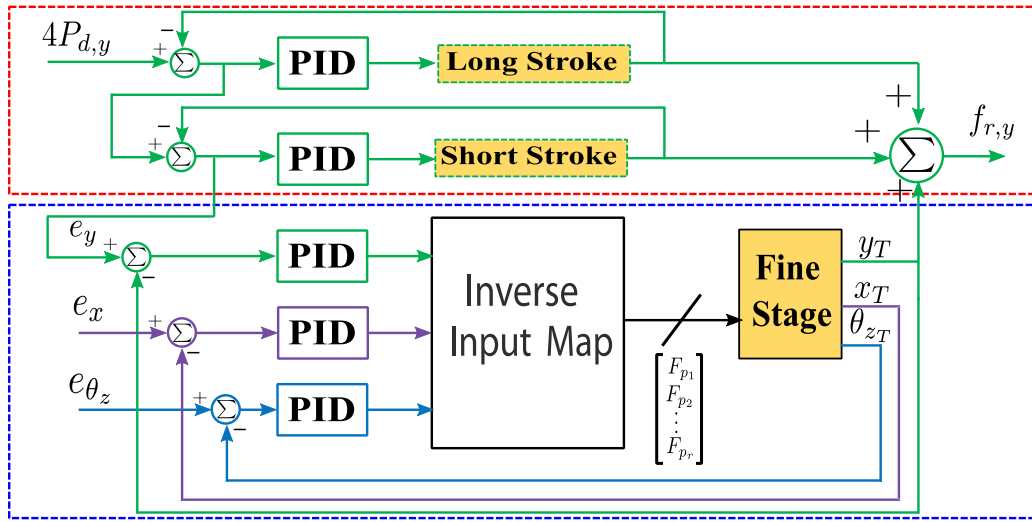


Fig. 13. The closed-loop position control of the reticle chain in the y -direction including the fine stage, where the LS and SS blocks represent the long-stroke and short-stroke models, respectively.

i.e., using all position information from reticle, optics and wafer chains, results in a better MA and MSD performance compared to the case when (4.8) is used even during the stepping phase. This is due to the additional information about the lens element, the mask and the wafer substrate motions, which is assumed to be available. Clearly, from Fig. 17, there is a possibility to increase the scanning and stepping speeds used as long as the fine stage working space limits are not exceeded. Therefore, introducing the fine stage as an add-on to an existing machine may allow increasing its throughput without altering its parameters or stability properties. Additionally, using the full chains synchronization allows less stringent requirements of the anti-vibration system at the machine frame, and therefore the optics chain need no more to be stationary. This will revolutionize the way the lithography machines are designed. Such advantages will also be subjects of future work.

6.2. Case 2

In this part, we present the simulation results of the same machine; however, now equipped with two fine stages, one at the reticle chain and the other at the wafer chain. The simulation results suggest that under the two fine stages, the machine performance depicted in Fig. 18 is quite similar to that captured in Fig. 15 which was obtained under one fine stage at the reticle chain when considering the synchronization error e_{syn} . Mainly, the differences are due to the additional dynamics introduced by the wafer chain fine stage, which under a better tuning can have a lesser deteriorating effect on the overall machine performance.

Interestingly, based on the fine stages interaction depicted in Fig. 9, these additional dynamics will also manifest themselves as an increase in the desired working space of the reticle chain fine stage that operates in the $\{x, y, \theta_z\}$ -directions as can be seen from Fig. 19. Fig. 20 shows the tracking performance of the reticle chain fine stage. In general, If a fine stage working space cannot be realized using the needed positioning technology, e.g. piezoelectric actuators, then the stiffnesses and masses values of stages, number and types of actuators, model assumptions and controlling technique used can be revised. This design process should be conducted in a cyclic fashion, accompanied by repetitive simulation of the fine stages using the herein developed machine model and repetitive verification using finite element analysis subjected to constrained parameters optimizations of the fine stages till the design meets the design requirements. Therefore, this study provides mainly one portion of the needed fine stages design cycle. As stated earlier, this design process will be thoroughly covered in separate studies.

The tracking performance of the fine stage at the wafer chain is captured in Figs. 21 and 22, which shows the benefit of this fine stage in leveling the wafer substrate when such level maps are available. Also, this stage can host active heat controlling systems, c.f. [44], that reduces the thermal loading on the wafer substrate during the exposure process.

7. Conclusion

In this investigation, a state-space motion related dynamical model of a typical lithography machine was presented in which the machine was divided into mainly three open-loop chains. By adopting a lumped mass-spring-damper approximation with noncolocated double-mass-block representation, the chains were subdivided into several stages which resulted in a straightforward mathematical modeling of the machine dynamics. After obtaining the kinematical features of the stages in each chain with respect to the floor frame, Newton's second law and Euler's equation were used to derive the stage dynamics. This hierarchical structure makes it possible to deal with more customized machines and their performances.

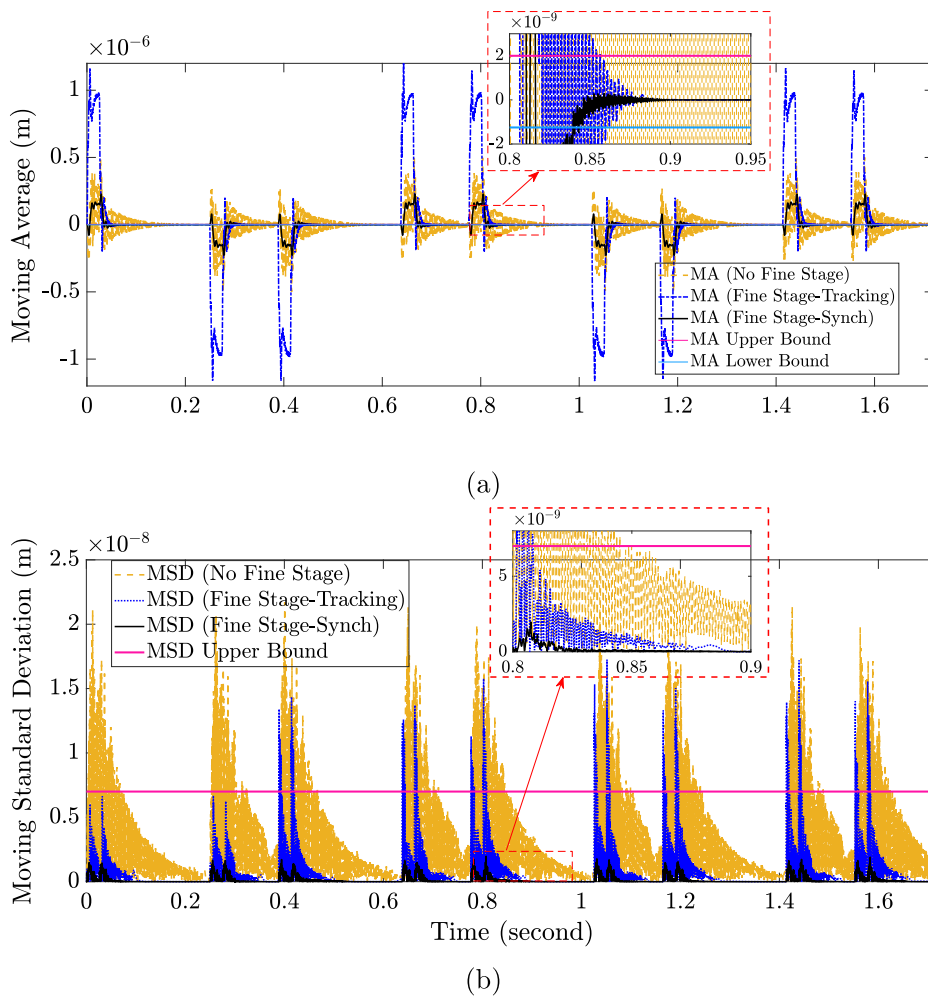


Fig. 14. The MA and MSD performance indices of the two machines with and without the reticle fine stage: (a) MA errors, and (b) MSD errors.

The dynamical model revealed the interactions among chains within the same machine where the trajectories followed by a chain end-effector can induce vibration and tracking errors in the other chains. Inside the optics chain, the internal dynamics were also captured when the optics box is approximated as a single thin lens element. Then, the ray-trace was obtained to capture the relation between chains interaction and optics chain during exposure phases. The developed model allows clear analysis and characterization of the machine such that the effect of each stage within a chain can be highlighted. Also, the overall machine performance can be assessed under the influence of the designed trajectories, controllers and vibration isolation subsystems in the three-dimensional space. This allows for better error budgeting and/or assigning sub-module specifications. Also, the developed model is more insightful regarding the control architecture in terms of global/local performance objectives. While focusing on lithography machines in this paper, a much broader audience can be approached when realizing that the modeling approach and steps that are taken also apply – for example – to electron microscopes or laser cutting machines.

To enhance machine performance, fine stages comprising multitudes of actuators were proposed. These fine stages offer the ability to improve the reticle and/or wafer stages performances in typical lithography machines, and they provide additional degrees of freedom to move the pattern and/or the wafer substrate in several directions. The effectiveness of these fine stages was assessed using the developed machine model where mainly synchronization errors were used. Also, using the developed machine model, the design requirements of the needed fine stages including mass, stiffness and working space dimensions can be obtained. In general, fine stages can be used with minimum modification to the available machine controllers with the possibility of increasing the machine throughput under optimized scan-and-step desired trajectories.

Acknowledgments

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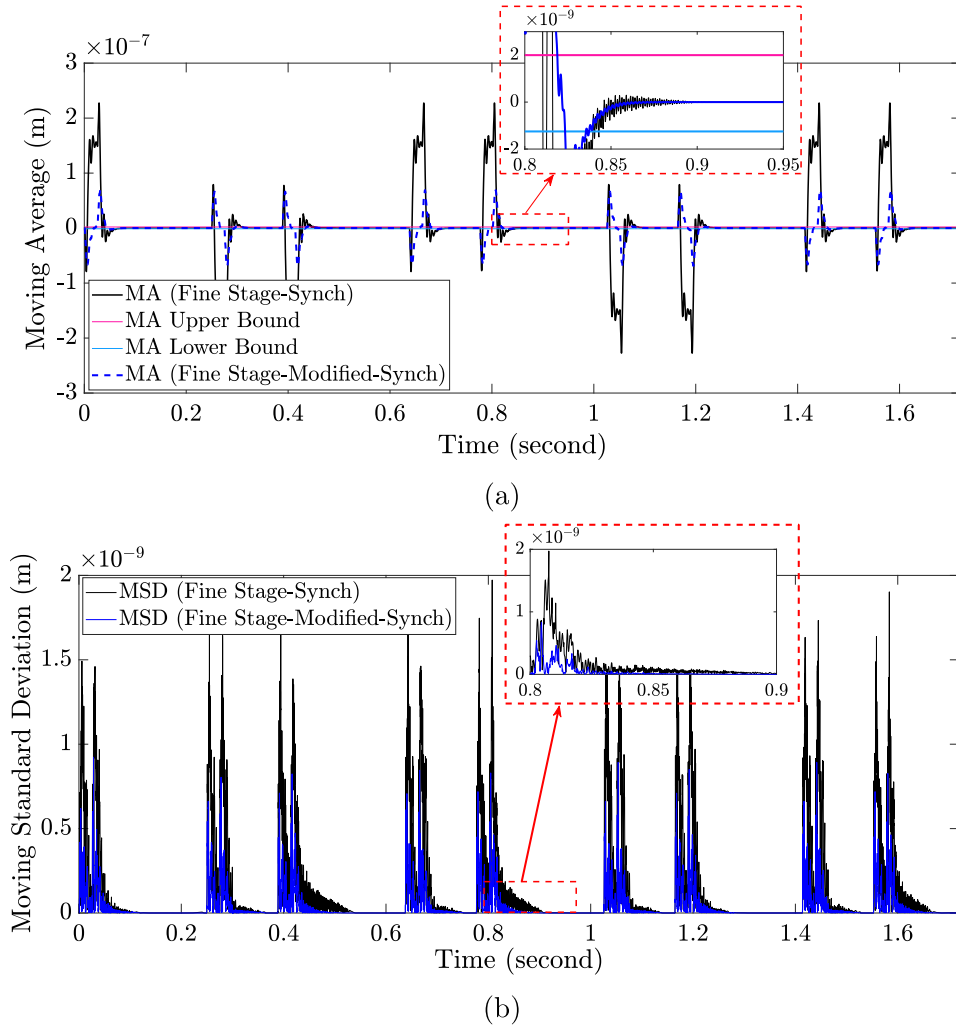


Fig. 15. The MA and MSD performance indices of the two machines with and without the reticle fine stage using the two versions of the synchronization error: (a) MA errors, and (b) MSD errors.

Appendix A. Matrices

The following matrices are needed when deriving the overall machine dynamical model in Section 2.2.

$$\begin{aligned}
 \bar{m}_{i_j} &= \sum_{k=i}^{N_j} m_{k_j} \\
 {}^0\tilde{\mathbf{C}}_{i_j} &= {}^0\mathbf{C}_{i_j} + 2m_{i_j} \begin{bmatrix} \omega_{(i-1)_j} \end{bmatrix} \\
 {}^0\tilde{\mathbf{K}}_{i_j} &= {}^0\mathbf{K}_{i_j} + {}^0\mathbf{C}_{i_j} \begin{bmatrix} \omega_{(i-1)_j} \end{bmatrix} + m_{i_j} \mathbf{B}_{(i-1)_j} \\
 {}^0\hat{\mathbf{C}}_{i_j,k} &= 2m_{i_j} \begin{bmatrix} \omega_{(k-1)_j} \end{bmatrix} \\
 {}^0\hat{\mathbf{K}}_{i_j,k} &= m_{i_j} \mathbf{B}_{(k-1)_j} \\
 {}^0\hat{\mathbf{C}}_{(i+1)_j} &= {}^0\mathbf{C}_{(i+1)_j} \\
 {}^0\hat{\mathbf{K}}_{(i+1)_j} &= {}^0\mathbf{K}_{(i+1)_j} + {}^0\mathbf{C}_{(i+1)_j} \begin{bmatrix} \omega_{i_j} \end{bmatrix} \\
 \mathbf{u}_{i_j}^0 &= {}^0\mathbf{R}_{i_j} \mathbf{u}_{i_j} \\
 \mathbf{m}_{i_j}^0 &= {}^0\mathbf{R}_{i_j} \mathbf{m}_{i_j}
 \end{aligned}$$

Appendix B. Machine parameters

The following parameters are needed to simulate the machine dynamics according to (2.12) and (2.14). The parameters are divided into subsections to ease accessing their numeric values.

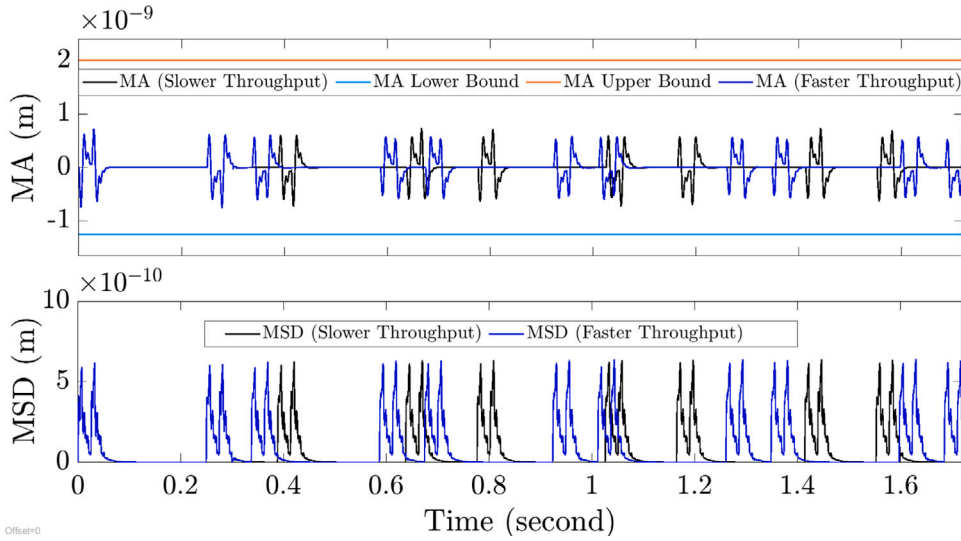


Fig. 16. The MA and MSD performance indices of the machine with the fine stage using the modified synchronization error (4.6) with and without increasing the throughput.

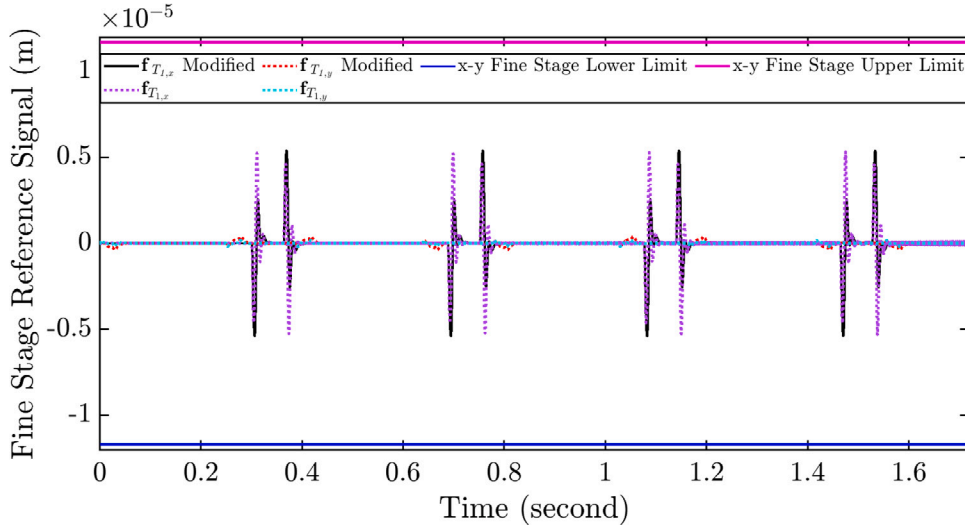


Fig. 17. Reference signal f_{T_i} of the reticle fine stage under the trajectories, shown in Fig. 12, given according to (4.7), i.e., modified, and (4.8).

B.1. Reticle and wafer chains common parameters

The following parameters are common to both the reticle and wafer chains depicted in Figs. 5 and 6, and also useful to understand the machine frequency allocation depicted in Fig. 11.

$$m_i = 0.5 \text{ kg}, i \in \{2, 4, 6\}$$

$$m_3 = 49 \text{ kg}, m_5 = 28 \text{ kg}, m_7 = 10.5 \text{ kg}$$

$${}^i\tilde{I}_i = 0.00400833 \text{ diag}([1, 1, 2]), i \in \{2, 4, 6\}$$

$${}^3\tilde{I}_3 = 4.10334 \text{ diag}([1, 1, 2])$$

$${}^5\tilde{I}_5 = 2.34477 \text{ diag}([1, 1, 2])$$

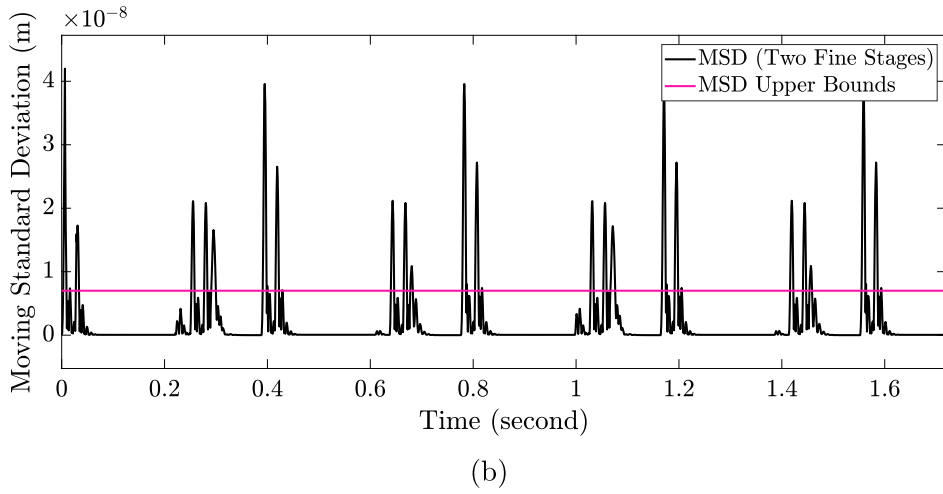
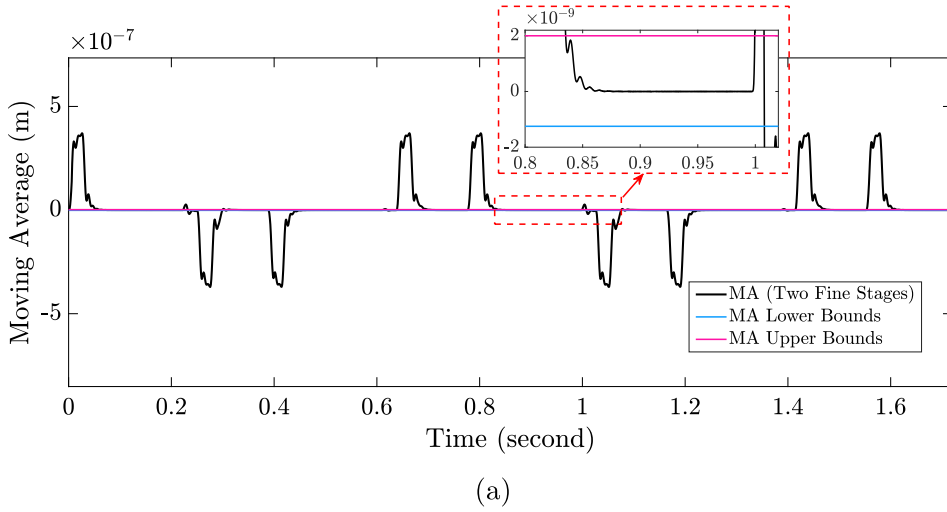


Fig. 18. The MA and MSD performance indices of the machine with the two fine stages depicted in Fig. 9: (a) MA errors, and (b) MSD errors.

$${}^7\tilde{I}_7 = 0.879287 \text{ diag}([1, 1, 2])$$

$$\mathbf{K}_i = 10^{12} \text{ diag}([1, 1, 1, 0.001, 0.001, 0.001]), i \in \{2, 4, 6\}$$

$$\mathbf{K}_3 = 10^9 \text{ diag}([0.01398, 0.01398, 0.01398, 1, 1, 1])$$

$$\mathbf{K}_5 = 10^9 \text{ diag}([0.01462, 0.01462, 0.01462, 1, 1, 1])$$

$$\mathbf{K}_7 = 10^9 \text{ diag}([0.01343, 0.01343, 0.01343, 1, 1, 1])$$

$$\mathbf{C}_i = 10^{10} \text{ diag}([1, 1, 1, 0.01, 0.01, 0.01]), i \in \{2, 4, 6\}$$

$$\mathbf{C}_3 = 10^8 \text{ diag}([0.0003701, 0.0003701, 0.0003701, 1, 1, 1])$$

$$\mathbf{C}_5 = 10^8 \text{ diag}([0.0002861, 0.0002861, 0.0002861, 1, 1, 1])$$

$$\mathbf{C}_7 = 10^8 \text{ diag}([0.0001679, 0.0001679, 0.0001679, 1, 1, 1])$$

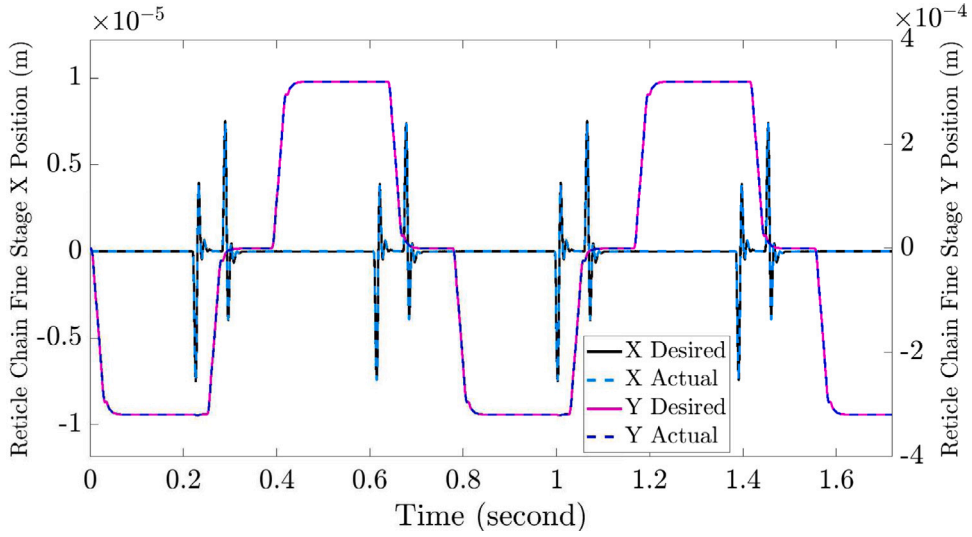


Fig. 19. Tracking of the reference signals $f_{T_{1,x}}$ and $f_{T_{1,y}}$, i.e. X Desired and Y Desired, of the reticle chain fine stage under the trajectories, shown in Fig. 12, given according to (4.10).

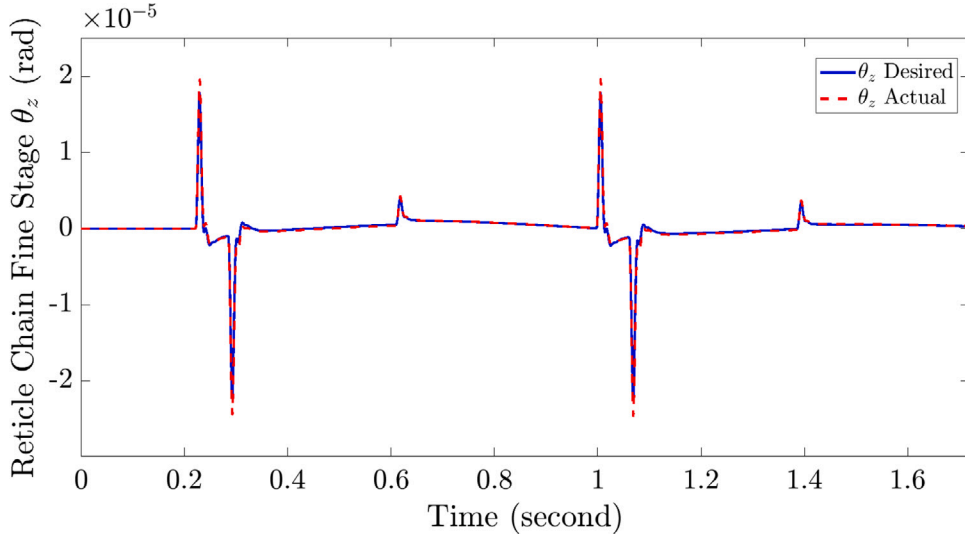


Fig. 20. Tracking of the reference signal $\theta_{T_{1,z}}$, i.e. θ_z Desired, of the reticle chain fine stage under the trajectories, shown in Fig. 12, given according to (4.11).

B.1.1. Reticle chain end-effector parameters

$$m_8 = 0.04 \text{ Kg}$$

$${}^8\tilde{I}_8 = 0.000133667 \text{ diag}([1, 1, 2])$$

$$\mathbf{K}_8 = 10^9 \text{ diag}([0.0063165, 0.0063165, 0.0063165, 1, 1, 1])$$

$$\mathbf{C}_8 = \text{diag}([710, 710, 710, 10^8, 10^8, 10^8])$$

B.1.2. Wafer chain end-effector parameters

$$m_8 = 0.2 \text{ Kg}$$

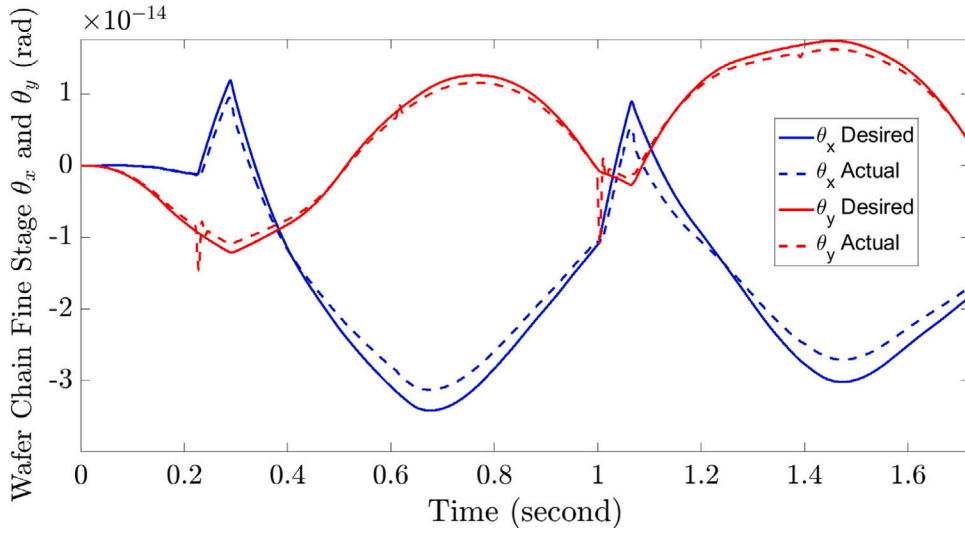


Fig. 21. Tracking of the reference signal $\theta_{T_{3x}}$ and $\theta_{T_{3y}}$, i.e. θ_x and θ_y Desired, of the wafer chain fine stage under the trajectories, shown in Fig. 12, given according to (4.13).

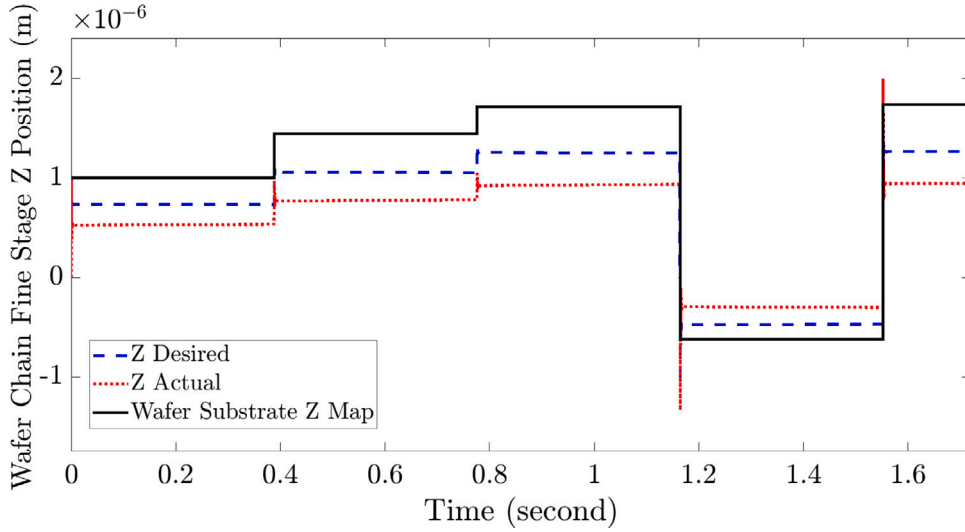


Fig. 22. Tracking of the reference signal $f_{T_{3z}}$, i.e. Z Desired, of the wafer chain fine stage under the trajectories, shown in Fig. 12, given according to (4.12).

$${}^8\tilde{I}_8 = 0.00512166 \text{ diag}([1, 1, 2])$$

$$\mathbf{K}_8 = 10^9 \text{ diag}([0.03158, 0.03158, 0.03158, 1, 1, 1])$$

$$\mathbf{C}_8 = 10^8 \text{ diag}([0.0000355, 0.0000355, 0.0000355, 1, 1, 1])$$

B.2. Optics chain parameters

The following parameters are used in the optics chains depicted in Fig. 7, and also useful to understand the machine frequency allocation depicted in Fig. 11.

$$m_2 = 520 \text{ kg}, m_3 = 200 \text{ kg}$$

$${}^2\tilde{I}_2 = \text{diag}([173.767, 293.367, 466.267])$$

$${}^3\tilde{I}_3 = \text{diag}([50.6667, 50.6667, 16])$$

$$\mathbf{K}_2 = 10^9 \text{diag}([0.4619, 0.4619, 0.4619, 1, 1, 1])$$

$$\mathbf{K}_3 = \text{diag}([6.396 \times 10^7, 6.396 \times 10^7, 6.396 \times 10^7, 20, 20, 20])$$

$$\mathbf{C}_2 = 10^8 \text{diag}([0.0069308, 0.0069308, 0.0069308, 1, 1, 1])$$

$$\mathbf{C}_3 = \text{diag}([1.5994 \times 10^5, 1.5994 \times 10^5, 1.5994 \times 10^5, 60, 60, 60])$$

B.2.1. Lens parameters

$$m_4 = 0.3 \text{ Kg}$$

$${}^4\tilde{I}_4 = 0.006 \text{diag}([1, 1, 1.8])$$

$$\mathbf{K}_4 = \text{diag}([1.07 \times 10^6, 1.07 \times 10^6, 1.07 \times 10^6, 20, 20, 20])$$

$$\mathbf{C}_4 = \text{diag}([800, 800, 800, 60, 60, 60])$$

B.3. Machine base-frame parameters

The following parameters are needed to obtain the machine base-frame dynamics in (2.4) and Section 2.1.

$$m_1 = 1400 \text{ Kg}$$

$${}^1\tilde{I}_1 = 10^3 \begin{bmatrix} 1.7618 & 0 & -0.799 \\ 0 & 3.9213 & 0 \\ -0.799 & 0 & 3.0933 \end{bmatrix}$$

$$\mathbf{K}_1 = 10^9 \text{diag}([0.00553, 0.00553, 0.00553, 1, 1, 1])$$

$$\mathbf{C}_1 = 10^8 \text{diag}([0.001244, 0.001244, 0.001244, 1, 1, 1])$$

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